

Hand Outline Color Modulates Perceived Effort while Lifting Real-World Objects in Mixed Reality

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Figure 1: Three experimental conditions: gray, red, and blue hand outlines, each displayed with medium line thickness. Red outlines produced greater perceived effort than gray outlines when participants lifted physical bottles.

Abstract

Although weight is grounded in physical forces exerted on the body, its perception also depends on visual information. We investigate whether visual augmentations applied directly to the hands can influence perceived weight and lifting effort when interacting with real-world objects whose physical properties remain unchanged. In a within-subjects study with 30 participants, we manipulated AR hand outline color (gray, red, and blue) and thickness (thin, medium, and thick). Participants experienced the nine resulting outline styles while lifting visually identical bottles of three physical weights, producing 27 outline trials, alongside three no-outline control trials. Outline color significantly affected perceived effort,

with red producing higher ratings than gray, despite no corresponding effect on perceived weight. Thickness did not reliably affect either measure. These findings suggest that visual augmentations to the body can selectively alter perceived physical exertion even when object mass and haptic feedback remain unchanged.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Empirical studies in HCI**; *Interaction techniques*.

Keywords

mixed reality, augmented reality, hand avatarization, hand outlines, perceived effort, weight perception, pseudo-haptics, embodied interaction

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1 Introduction

Humans continuously make judgments about the physical properties of objects in their environment. Among these properties, weight plays an important role in how we interact with the physical world, informing how we grasp, lift, carry, and manipulate everyday objects. Our expectations about an object's weight can influence how we prepare to interact with it, while the forces experienced during interaction allow us to continually update these expectations [26]. In this way, weight perception forms part of the broader relationship between humans and our physical environments.

Although weight is ultimately grounded in the physical forces exerted on the body, its perception does not arise from haptic information alone. Human perception integrates information across multiple sensory modalities [16, 17], and visual cues can influence how physical properties such as size, material, shape, and weight are perceived [8, 18]. For example, judgments of an object's weight can be derived from its visual appearance, such as its size [10] and surface color [41]. Head-mounted displays provide a particularly interesting means of manipulating these visual cues. Virtual reality (VR), augmented reality (AR), and mixed reality (MR) systems can selectively alter how users perceive their bodies and the objects with which they interact, while preserving the user's underlying physical movements.

Prior work has leveraged visual manipulations in VR and AR to influence haptic perception. For instance, Baek et al. [3] manipulated the line style and color of outlines surrounding users' virtual hands during interactions with vibrotactile virtual objects in VR. They found that line style significantly modulated perceived roughness, arousal, and valence, with sharp outlines increasing perceived roughness and arousal, while color primarily influenced valence. Similarly, David et al. [13] applied visual motion and color cues directly to users' embodied virtual hands in AR while they interacted with a visually neutral virtual surface. Their visual cues systematically shifted affective responses, although they did not reliably induce sustained tactile or thermal sensations.

These works demonstrate the potential for visual cues applied directly to the hand to shape tactile and affective experience. However, both examined interactions with *virtual* objects or surfaces, leaving unclear whether similar hand augmentations can influence haptic perception when interacting with *real-world* physical objects whose inherent physical properties remain unchanged. This distinction is important. Unlike virtual objects, real objects provide fixed physical cues, including mass and the forces exerted on the body during lifting and manipulation. MR therefore provides a means of testing whether visual cues can shift haptic judgments even when they conflict with, or augment, veridical physical feedback. Additionally, the effects of such hand augmentation on dimensions of haptic perception other than roughness, temperature, and valence-arousal are underexplored in the literature. Weight offers a particularly interesting case because it is directly grounded in physical force while also being susceptible to visual expectations. Examining weight perception therefore allows us to investigate how far MR-based visual cues can influence judgments of physical objects despite the continued presence of real-world haptic feedback.

Based on this research gap, we formed the following research questions:

RQ1. How do MR hand outline styles (*thickness and color*) influence **weight perception** of real-world objects of *differing weights*?

RQ2. How do MR hand outline styles (*thickness and color*) influence **perceived effort** required to lift real-world objects of *differing weights*?

In this work, we examine whether the thickness and color of hand outlines influence users' judgments of the weight of real physical objects. In parallel, we investigate whether these visual manipulations influence the perceived effort required to lift the object, allowing us to examine whether changes in visual representation affect not only judgments about the object, but also users' perception of their own physical exertion.

To investigate these effects, we conducted a user study manipulating two visual properties of hand outlines: line thickness and line color, each across three levels. Participants lifted three real-world physical objects of differing weights under each visual condition, alongside control trials in which no hand outline was presented. Following each interaction, participants reported their perception of the object's weight and the effort required to lift it. This study builds on prior research by focusing on physical objects rather than virtual objects, and examining perceived weight and lifting effort rather than roughness, temperature, and valence-arousal.

Our findings contribute to the design of mixed and augmented reality experiences in which virtual content and physical objects coexist. Understanding how visual representations of the body influence physical perception informs applications such as mixed reality tabletop games, training and education, and pervasive extended reality experiences, where users routinely interact with real objects while receiving digitally mediated visual feedback. We conclude by discussing the implications of these findings for the design of such interactions.

2 Related Work

In this section, we review prior research on weight perception of objects, modulating haptic perception in XR using multisensory cues, and augmented reality hand avatarization.

2.1 Haptic Perception of Real Objects

People integrate multiple senses to understand an object's physical properties. Ernst and Banks [16] found that people rely more on whichever sense provides clearer information. For example, when visual information is blurry or unreliable, people rely more on touch. When visual information is clear, they rely more on vision. For surface texture, touch often has a stronger influence than vision because it provides direct information about properties such as roughness [20]. However, visual information can still change how a surface feels. Suzuishi et al. [39] found that watching a moving sine wave grating altered the perceived roughness of real abrasive surfaces while participants touched them. Similarly, Ujitoko et al. [40] found that participants perceived the same vibration-based texture as rougher when its visual pointer moved back and forth.

Similarly to other haptic properties, the perceived weight of an object is not determined by its mass alone. The size-weight illusion shows that the smaller of two objects of equal mass is consistently judged to be heavier [10]. The illusion persists after

explicit knowledge that the objects weigh the same, as well as after the sensorimotor system (specifically the grip and load forces used to lift the objects) has adapted to their true mass [18, 34]. This shows that weight perception of objects is not a direct readout of the physical forces acting on the hand, but a cognitive construction shaped by expectations and integration of multisensory cues.

These perceptual expectations are heavily informed by visual appearance. For instance, visual cues regarding an object's material properties significantly alter both the anticipatory lifting forces applied by users and their subsequent judgments of heaviness [11]. Furthermore, the expectations underlying these judgments are highly malleable, shifting when users repeatedly experience objects that violate natural size-weight correlations [19].

Surface color serves as another powerful visual cue. While darker objects are judged heavier than lighter ones based on vision alone, this association inverts once the objects are physically lifted. This is a phenomenon known as the brightness-weight illusion [41]. This pattern closely mirrors the size-weight illusion, suggesting a deeply embedded, cross-sensory correspondence where heaviness aligns with darkness independent of an object's physical dimensions [42].

Together, these works show that haptic perception, including weight perception, is a multisensory experience which can be modulated by visual cues grounded in the physical world. The following section explores how digitally mediated visual imagery can modulate haptic perception, through the use of XR.

2.2 Object-Centered Visual Modulation in XR

XR systems can visually alter real objects without changing their physical properties, allowing researchers to examine how digitally mediated appearance shapes haptic perception. Iesaki et al. [24] superimposed virtual texture imagery onto physical objects in MR and found that this changed participants' impressions of roughness. Similarly, Punpongson et al.'s SoftAR [35] projected simulated surface deformation onto soft physical objects, allowing their apparent softness to be adjusted independently of their actual compliance. These findings show that object-centered visual augmentation can alter perceived material qualities even when users remain in contact with a real surface.

Similar methods have targeted perceived weight. Ban et al. modified the apparent brightness of handled objects in AR, influencing lifting behavior and reducing unnecessary muscular effort and fatigue [5]. Rohrbach et al. [37] overlaid equally sized virtual cubes onto physical cubes of different sizes but identical mass. This eliminated the usual size-weight illusion during initial lifts and resulted in more similar anticipatory fingertip forces. Hashiguchi et al. [22] instead altered the apparent length of a physical rod through AR and diminished reality, finding that visually longer rods were perceived as lighter. Their later work confirmed this relationship and connected it to changes in the perceived center of gravity, while showing that simply hiding part of the rod did not reliably change its perceived weight [21].

Together, these works demonstrate that visual augmentations applied to objects can influence the perceived roughness, softness, and weight of real physical stimuli. However, they predominantly modify the visual representation of the object itself. This leaves

open whether haptic perception of a real object can also be modulated by applying a lightweight visual cue to the user's body while leaving the object's appearance unchanged. The following section reviews visual effects applied directly to the represented hand.

2.3 Hand-Centered Visual Effects in XR

Visual feedback in XR can be applied directly to the user's represented hand to influence haptic perception without modifying the physical stimulus itself. Early pseudo-haptic work demonstrated this using spatial and morphological changes to the hand. Pusch et al. [36] dynamically displaced the visual representation of users' hands in video see-through AR, producing distinguishable sensations of virtual force. Ban et al. [6, 7] similarly manipulated the appearance of users' hands while they interacted with physical objects, showing that hand deformation could alter perceived object size [6] and stiffness [7]. Sato et al. [38] later applied visual effects to a projected virtual hand to produce impressions of unevenness, slipperiness, and softness without physical haptic feedback. More recently, Yamamoto et al. [43] used virtual finger deformation to visually represent force, finding that it improved users' ability to discriminate weight and force differences.

Baeck et al. [3] explored a less intrusive form of hand-centered feedback by applying visual effects to the outline of an embodied virtual hand. They varied outline shape, motion, and color while participants interacted with virtual objects accompanied by vibrotactile feedback. Sharp outlines increased perceived roughness and arousal, while color primarily affected valence. In subsequent work [4], Baeck et al. used similar hand-outline cues to visually communicate different levels of virtual-object roughness between collaborators. Their results demonstrate that lightweight, hand-anchored outlines can both influence and communicate haptic properties without modifying the visual appearance of the object itself. David et al. [13] similarly applied motion and color effects to users' tracked hands in MR while leaving the contacted virtual surface visually neutral. These effects systematically influenced affective responses, although they rarely produced sustained tactile or thermal sensations.

Prior work therefore demonstrates that visual effects applied to the hand can shape haptic experience. However, these effects have primarily been examined with virtual objects, simulated haptic feedback, or substantial spatial deformation of the represented hand. Comparatively little is known about whether lightweight visual effects applied to the hand can alter perception when interacting with real objects that continue to provide veridical physical feedback. We address this gap in the current work.

2.4 Color and Visual Weight for Affective User Experiences

Color can shape affective responses, motivation, and users' interpretation of visual information. A large body of color-psychology research suggests that these effects are systematic, although they depend on both the properties of the color and the context in which it appears [14, 27]. In particular, red is frequently associated with heightened arousal, warning, and avoidance-related responses. For example, Elliot et al. found that exposure to red in achievement contexts elicited avoidance motivation and impaired subsequent task

performance [15]. In contrast, blue and green are more commonly associated with pleasant and positive, lower-arousal affect [9, 27]. Briki and Hue directly compared red, blue, and green and found that red produced greater arousal and dominance, whereas blue and green were perceived as more pleasurable [9]. These associations can also influence interactive experiences. Mehta and Zhu showed that red and blue induce different motivational orientations, with red promoting avoidance-oriented processing and blue promoting approach-oriented processing [33]. In HCI, website color has likewise been shown to affect perceived trust and satisfaction [12].

Beyond color, basic graphical properties can carry connotative meaning. This relationship is particularly established in typography, where visually bold forms are described as heavy and delicate forms as light. Lewis and Walker found that observers reliably distinguished typefaces along a heavy-light dimension and responded more slowly when the connotative weight of a typeface conflicted with the heaviness of the concept being judged [32]. Henderson et al. similarly identified weight as one of the underlying dimensions structuring impressions of typeface design [23]. In packaging research, delicate and bold typefaces have consequently been used as nonverbal cues for lightness and heaviness, respectively [29].

Together, the above works suggest that thicker, visually weightier marks can evoke heaviness-related associations, and that color can be an affective cue capable of altering users' subjective experience of an interaction. However, it remains unknown whether these associations transfer from static graphical forms to haptic judgments when an outline is attached to an embodied hand and the physical properties of the lifted object remain unchanged.

3 Study Design

To address our research questions on the effect of hand outline styles on perceived weight (**RQ1**) and effort (**RQ2**), we conducted a $3 \times 3 \times 3$ within-subjects study manipulating outline thickness, outline color, and physical bottle weight. The factorial portion of the study contained 27 conditions, comprising all combinations of three outline thicknesses (thin, medium, thick), three outline colors (gray, red, blue), and three bottle weights (328g, 428g, 528g). We additionally included a no-outline control condition for each bottle weight, resulting in 30 trials per participant. Because the control condition contained neither an outline color nor an outline thickness, we treated it as a separate baseline for descriptive purposes rather than as an additional level of either factor. Figure 2 shows the visual conditions used in the study. This section describes our participants, variables, procedure, and hypotheses.

3.1 Participants

Thirty-two participants were screened, enrolled, and completed the study. Two participants were excluded prior to outcome analysis and replaced to maintain the planned analyzed sample of 30 participants and the intended counterbalancing scheme. One participant was inadvertently assigned a Williams-square trial order that had already been used by an earlier participant, resulting in a duplicate counterbalancing sequence. A second participant was excluded because they did not engage with the study procedures as instructed, resulting in responses that were not considered valid for analysis. The final analyzed sample therefore consisted of 30 participants.

Participants in the analyzed sample were between 19 and 41 years of age ($M = 26.13$, $SD = 5.56$). Fifteen participants identified as men (50.0%), 13 as women (43.3%), and two as nonbinary (6.7%). Eleven participants (36.7%) identified as South Asian, seven (23.3%) as East Asian, four (13.3%) as Middle Eastern or North African, two (6.7%) as Black or African or Afro-descendant, one (3.3%) as White or European, and one (3.3%) as Mixed or Multiracial. Three participants (10.0%) selected another ethnicity, of whom two specified Southeast Asian, and one participant (3.3%) preferred not to report their ethnicity.

Participants generally had limited prior experience with head-mounted displays. Eleven participants (36.7%) reported never having used one, 10 (33.3%) had used one once or twice ever, eight (26.7%) reported using one once or twice per year, and one (3.3%) reported using one once or twice per month.

Participants were required to be between 18 and 64 years old, be able to read, write, speak, and understand English, have normal or corrected-to-normal vision, be able to raise both arms and move their head freely, and have no severe history of cybersickness or motion sickness. All participants provided informed consent before participating.

All participants provided informed consent before participating. Pilot and test trials were also excluded from the analysis. This study received ethics approval by our Institutional Review Board (Protocol Number:*****)

3.2 Independent Variables

Line thickness. We manipulated the thickness of the hand outline at three levels: *thin* (1.5 mm), *medium* (4.0 mm), and *thick* (8.0 mm). These values correspond to world-space outline widths of 0.0015 m, 0.0040 m, and 0.0080 m, respectively. The levels were selected to provide three clearly distinguishable degrees of visual outline prominence. Their spacing was tuned perceptually so that the difference between thin and medium appeared comparable to the difference between medium and thick when viewed at approximately arm's length; consequently, the levels are not evenly spaced in metric units. This manipulation extends prior work showing that visual properties of hand outlines can modulate haptic and affective perception [3].

Line color. We manipulated outline color using *gray*, *red*, and *blue*. Their sRGB values were (0.430, 0.430, 0.430), (0.850, 0.100, 0.100), and (0.100, 0.402, 0.900), respectively. All three colors were matched to the same relative luminance ($Y = 0.155$), allowing us to vary hue while controlling for luminance. We selected red and blue because prior work on hand-outline feedback has associated red with lower valence and greater arousal, urgency, or irritation, whereas blue is generally associated with a calmer and more pleasant response [3]. We used gray as an achromatic neutral comparison. Importantly, previous work has shown that the apparent heaviness associated with a color increases with chroma and decreases with lightness, whereas hue itself has comparatively little effect [2]. We therefore avoided using white or black as the neutral condition and luminance-matched all three colors to reduce the possibility that differences in perceived weight could be attributed to brightness rather than hue.

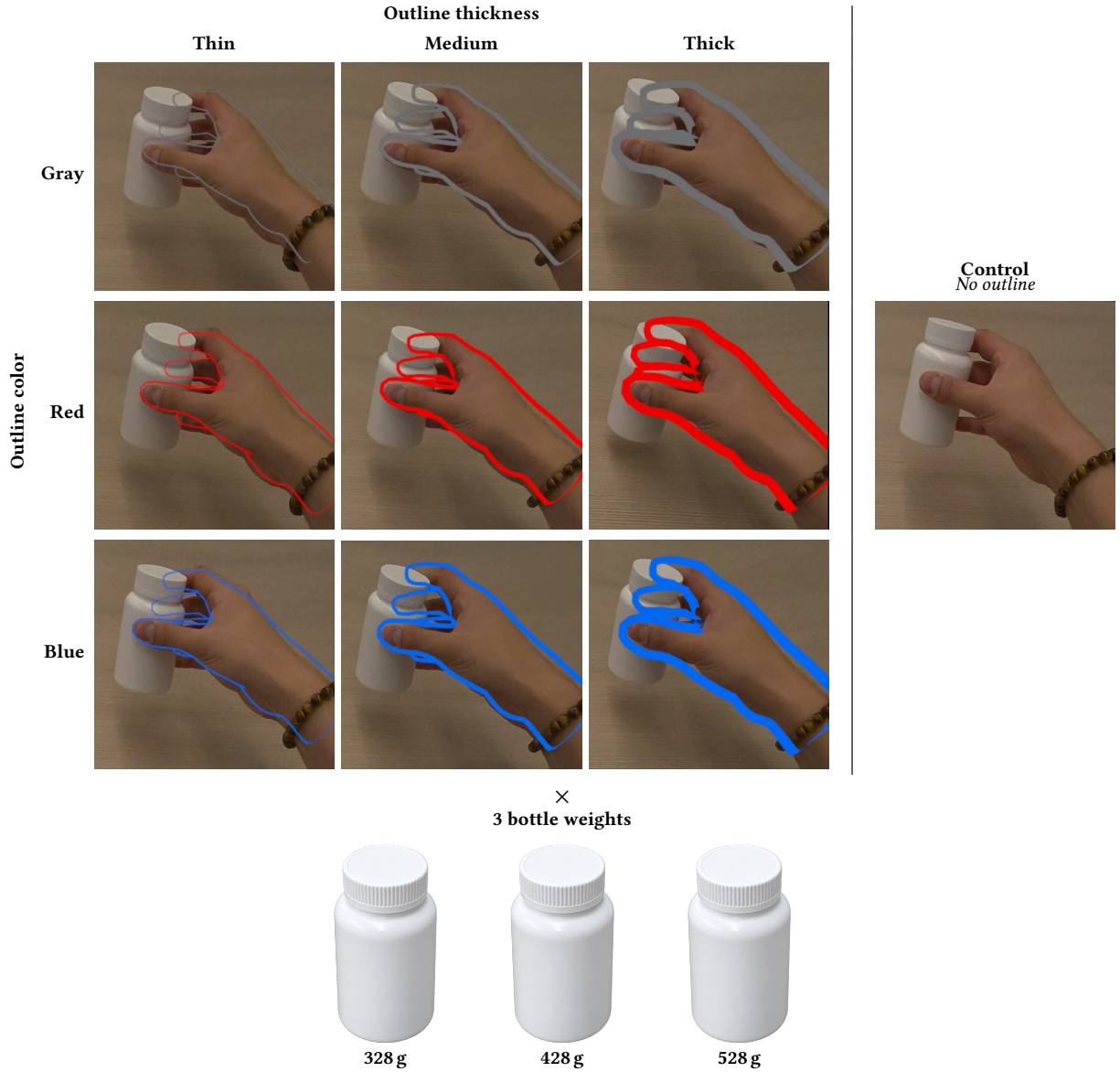


Figure 2: Experimental conditions. The study manipulated outline color (gray, red, blue), outline thickness (thin, medium, thick), and bottle weight (328 g, 428 g, 528 g). The visual manipulation consisted of nine outline conditions plus a no-outline control condition. Each of these 10 visual conditions was tested with all three bottle weights, resulting in 30 total trials.

Bottle weight. Participants lifted bottles weighing 328 g, 428 g, and 528 g, with 100 g separating adjacent levels and 200 g separating the lightest and heaviest bottles. Physical weight was varied for two reasons. First, presenting multiple physical weights reduced the likelihood that participants would learn that every trial involved an identical object weight and simply repeat the same judgment across trials. Second, including physical weight as an experimental factor allowed us to test whether any effect of hand-outline appearance depended on the actual weight being lifted. The weight manipulation additionally served as a manipulation check, as participants

should reliably distinguish the lightest and heaviest bottles if the perceived-weight measure is sensitive to physical mass.

3.3 Dependent Variables

Perceived weight. Following each lift, participants responded to the question “How heavy did the bottle feel?” using an integer slider ranging from 1 (*Not at all heavy*) to 100 (*Extremely heavy*). We adopted the 1–100 response format used by Baeck et al. [3] to measure haptic judgments under hand outline manipulations.

Perceived effort. Participants also responded to the question “How much effort did lifting the bottle require?” using an integer slider

ranging from 1 (*No effort at all*) to 100 (*Maximum effort*). Both questions were presented simultaneously on the same panel and were oriented in the same direction, with larger values representing greater perceived weight or effort.

For both perceived weight and perceived effort, the slider was reset to its midpoint (50) before every trial. Participants were required to interact with both sliders before the trial could be submitted, preventing unanswered items or responses inherited from a previous trial.

Open-ended responses. We asked participants open-ended questions at the end of the study to obtain qualitative explanations supporting any effects observed in our quantitative analysis. Table 1 outlines the open-ended post-study questionnaire.

Table 1: Open-ended post-study survey questions.

Q1.	Did the AR hand outlines influence your perception of the physical properties of the bottle? Explain your answer.
Q2.	Did the thickness (thin / medium / thick) of the hand outline influence how heavy the bottle felt, or how easy to lift it felt? Explain why.
Q3.	Did the color (gray / red / blue) of the hand outline influence how heavy the bottle felt, or how easy to lift it felt? Explain why.

3.4 Procedure

Before beginning the experimental trials, participants completed the pre-study screening and consent procedure and put on the Meta Quest 3 head-mounted display. The experimenter entered an anonymized three-digit participant code and participant number into the experiment software. The participant number determined the participant's trial order.

Each participant completed all 30 conditions exactly once. Trial order was counterbalanced using a Williams square of order 30. In addition to balancing the position in which each condition occurred, the Williams design balanced first-order carryover between conditions, reducing the possibility that a rating was systematically influenced by the immediately preceding outline or bottle condition. All 30 conditions were counterbalanced together rather than grouping trials into separate bottle-weight blocks.

At the beginning of each trial, the experimenter initiated the next condition and placed the required bottle in front of the participant. The participant then grasped and lifted the bottle. In outline conditions, the specified outline appeared around the participant's hands when the system detected contact with the bottle and remained visible while the bottle was held. In control trials, participants performed the same interaction without a hand outline. Participants then returned the bottle to the table.

After the lift was complete, the experimenter removed the bottle and presented the response panel in front of the participant. Participants rated both the perceived weight of the bottle and the effort required to lift it using direct fingertip interaction with the two 1-100 sliders, and then submitted their responses. The response panel was removed before the next trial began. Lifts shorter than

0.4 s were treated as accidental contact or a fumble rather than as a completed trial and did not proceed to the rating stage. This procedure was repeated until all 30 conditions had been completed. After the final trial, participants completed the post-study questionnaire shown in Section 3.3.

3.5 Hypotheses

Prior research has demonstrated that visual effects applied directly to hand outlines can alter haptic and affective judgments, with more intense outline styles increasing perceived roughness and arousal and red outlines producing more negative affect than other colors [3]. Separate research on visual and typographic weight indicates that bold, visually weighty forms can convey heaviness-related meaning [23, 29, 32]. We therefore expected that increasing the visual weight of the hands through a thicker outline could transfer this association to the concurrent lifting experience. Extending these findings from static graphical stimuli and virtual haptic interactions to the lifting of real physical objects, we formed the following hypotheses:

- H1.** Thicker outlines will increase the perceived weight of real-world objects compared to thinner outlines.
- H2.** Red outlines will increase the perceived effort required to lift real-world objects compared to blue outlines, with gray outlines falling between the two.

4 System Description

Participants wore a Meta Quest 3 connected by USB cable to a laptop with an Intel Core i9-11900H processor and 32 GB of memory, running Windows 11. The outline around the participant's hands appears when they take hold of the bottle and stays for as long as they hold it. We use computer vision to detect when participants grab the object. The headset streamed its passthrough camera view to the laptop at 15 frames per second and the laptop did all of the analysis.

Grabbing detection. We used two off-the-shelf models, both running on the CPU. YOLOv8s, a general-purpose object detector, located the bottle using its existing bottle category with no additional training, and MediaPipe's hand landmark model returned the positions of the finger joints. A hand counted as touching the bottle when at least three of its five fingertips fell inside the detected outline of the bottle, widened by 15% to allow for the part of it the hand covers. A lift began on the first frame of contact, because an outline that appears after the lift has started is not present for the moment it is meant to influence. A lift ended only after 18 consecutive frames without contact, about 1.2 seconds, because a hand briefly obscuring the bottle mid-lift would otherwise read as putting it down.

Hand Outlines. While the bottle is held, both of the participant's real hands are outlined in the headset. The outline is drawn on the headset from the hand model that the Quest's hand tracking already provides, so it follows the real hand smoothly. We applied a shader to the hand model, which used the inverted hull method to create an outline around it, by expanding a copy slightly outward in every direction, and drawing only the parts of the copy that fall outside the hand's own silhouette, which leaves an even band

around the hand that the hand itself occludes. The width of that band is the thickness manipulation.

Bottles. For the physical objects involved in the task, we chose opaque white pill bottles, which we deemed to be a neutral and generalizable object to test our hypotheses. The pill bottles were each 2.32 inches wide and 4.13 inches high, with capacities of 200ml. We controlled the weights of our three bottles by filling them with varying ratios of 100g metal weights and lightweight polystyrene filler to keep the weights locked in place. After weighing each bottle on a digital scale, their weights came out to 328g, 428g, and 528g. All three bottles were visually identical.

5 Results

5.1 Analysis Method

We analyzed perceived weight and perceived effort separately using $3 \times 3 \times 3$ repeated-measures ANOVAs with outline thickness (*thin*, *medium*, *thick*), outline color (*gray*, *red*, *blue*), and bottle weight (328 g, 428 g, 528 g) as within-subjects factors. When Mauchly's test indicated a violation of the sphericity assumption, we report Greenhouse-Geisser-corrected degrees of freedom and p -values. For significant omnibus effects, we conducted pairwise comparisons with Bonferroni correction. We report partial eta squared (η_p^2) as the effect-size estimate. The no-outline control was not included as a level of color or thickness because it contained neither manipulation; it is included descriptively in Figures 3 and 4.

5.2 Effects of Hand Outline Color

Outline color did not significantly affect perceived weight, $F(2, 58) = 1.46$, $p = .240$, $\eta_p^2 = .048$. Mean perceived-weight ratings were similar across gray ($M = 54.84$, $SE = 2.86$), red ($M = 55.93$, $SE = 2.96$), and blue ($M = 55.17$, $SE = 2.99$) outlines. These mean ratings are visualized in Figure 3A.

We found a significant main effect of outline color on perceived effort, $F(2, 58) = 4.27$, $p = .019$, $\eta_p^2 = .128$. Bonferroni-corrected pairwise comparisons showed that participants reported significantly greater effort with red outlines ($M = 49.48$, $SE = 3.07$) than with gray outlines ($M = 47.84$, $SE = 3.07$), with a mean difference of 1.64 points, $SE = 0.60$, 95% CI [0.10, 3.17], $p_{\text{Bonf.}} = .034$. Blue outlines ($M = 48.43$, $SE = 3.00$) did not significantly differ from either gray ($p_{\text{Bonf.}} = .815$) or red ($p_{\text{Bonf.}} = .232$). Thus, the effect of color on perceived effort was driven by greater effort ratings for red relative to gray, with blue falling numerically between the two. Figure 3B visualizes this effect.

We found no significant interactions involving outline color for either perceived weight or perceived effort, indicating that the observed color effect on effort did not significantly depend on outline thickness or bottle weight.

5.3 Effects of Hand Outline Thickness

We found no significant main effect of outline thickness on perceived weight, $F(2, 58) = 0.94$, $p = .397$, $\eta_p^2 = .031$. Mean perceived weight increased slightly across the thin ($M = 54.68$, $SE = 2.92$), medium ($M = 55.36$, $SE = 2.94$), and thick ($M = 55.90$, $SE = 3.01$) conditions, but these differences were not statistically significant. These ratings are visualized in Figure 4A for descriptive purposes.

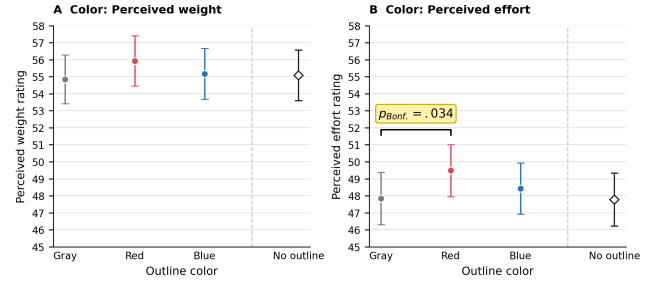


Figure 3: Mean perceived weight and effort ratings by outline color, averaged across line thicknesses and object weights. Error bars represent half of one standard error. The no-outline control is included descriptively. Red outlines produced significantly greater perceived effort than gray outlines ($p_{\text{Bonf.}} = .034$).

Similarly, outline thickness did not significantly affect perceived effort. Because the sphericity assumption was violated for this factor, we report the Greenhouse-Geisser-corrected result, $F(1.59, 46.00) = 0.52$, $p = .559$, $\eta_p^2 = .017$. Mean effort ratings were comparable across thin ($M = 48.43$, $SE = 3.02$), medium ($M = 48.22$, $SE = 3.00$), and thick ($M = 49.11$, $SE = 3.19$) outlines. Bonferroni-corrected comparisons between thickness levels were also nonsignificant. These ratings are visualized in Figure 4B for descriptive purposes. We additionally found no significant interactions involving outline thickness for either dependent variable. Overall, we found no evidence that increasing outline thickness systematically altered either perceived weight or perceived effort.

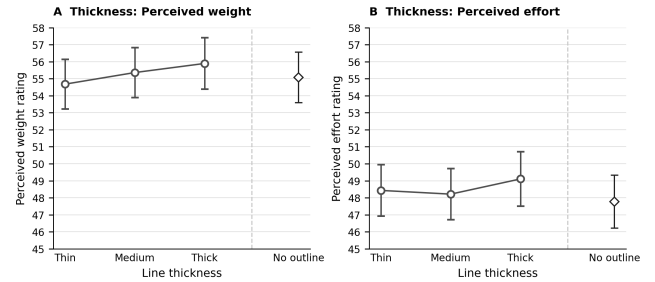


Figure 4: Mean perceived weight and effort ratings by outline thickness, averaged across outline colors and object weights. Error bars represent half of one standard error. The no-outline control is included descriptively.

5.4 Effects of Bottle Weight

As expected, bottle weight significantly affected both perceived weight, $F(2, 58) = 78.75$, $p < .001$, $\eta_p^2 = .731$, and perceived effort, $F(2, 58) = 75.69$, $p < .001$, $\eta_p^2 = .723$. Ratings increased with physical bottle weight for both measures, and all Bonferroni-corrected pairwise comparisons between the 328g, 428g, and 528g bottles were significant ($p_{\text{Bonf.}} < .001$).

6 Discussion

We interpreted our results through the key findings presented below, revisiting our hypotheses and the extent to which they were supported. We drew on responses from the open-ended questionnaire to provide additional context for participants' experiences and interpretations of the visual manipulations. We also discuss applications of our findings to various domains, limitations of this work, and future directions.

6.1 Key Findings

Outline Color Modulates Effort, but not Weight Perception.

We observed a significant effect of hand outline color on effort ($p = .019$), with a medium effect size ($\eta_p^2 = .128$). In contrast, neither outline color nor outline thickness significantly affected perceived weight. This pattern suggests that there is a distinction between the perceived weight of objects, and the perceived effort that is required to lift them up. Open-ended responses provided some context for this distinction. Several participants reported focusing primarily on the physical weight of the bottles and paying comparatively little attention to the hand outlines:

"I was only focusing on the weights and even though the outlines seemed to change in color and thickness, I didn't pay any attention to it." (P10)

These responses suggested that perceived weight may have been anchored more strongly to the physical sensations produced by lifting the objects, limiting the influence of the visual manipulation. Another participant stated that the hand outline color acted as a psychological cue, without changing the physical sensation itself:

"Color seemed to act as a psychological cue for heaviness rather than changing physical sensation." (P20)

This interpretation aligns with research by David et al. [13], which found that visual and auditory input did not induce haptic illusions of tactile and thermal properties, but they instead impacted emotional aspects of the interaction, such as the valence-arousal scale. Our results similarly suggest that perceived effort is tied closely with perception of one's own physical abilities, rather than the physical properties of the objects, and is therefore more influenced by shifts in affective perception. This would explain why we found color to modulate effort, but not perceived weight, which seems to have been judged more objectively. This dissociation between effort and weight ratings suggests that AR visual augmentations may influence the experience of physical interaction without necessarily altering perception of the object's physical properties.

Red Outlines Increase Effort in Comparison to Gray Outlines. Bonferroni pairwise comparisons further showed that perceived effort was significantly higher with red outlines than with gray outlines ($p = .034$). Blue outlines descriptively fell between red and gray, but did not significantly differ from either. Our hypothesis (H2) predicted that red outlines would increase the perceived effort required to lift real-world objects compared to blue outlines, with gray landing somewhere in the middle. Although our results do show that red outlines yielded significantly higher effort than gray outlines, it is interesting that blue and gray are swapped. Instead of gray landing in the middle and acting as a neutral condition, blue

has taken that place instead. Participants tended to agree with this ordering:

"Red felt heavier. Gray felt lighter and blue in the middle." (P4)

"Yes, I think gray color made them feel the lightest, and then blue and then red made them feel the heaviest." (P15)

Therefore, H2 is partially supported by our results. We posit that this ordering may not be the result of a single reason, but a combination of multiple factors. One possible explanation for the higher effort ratings under red outlines is the association of red with intensity and heightened arousal, which formed the basis for H2 and has also been observed in prior work with virtual hand outlines [3]. Some participants similarly described red as making the interaction feel more effortful:

"The colour of the hand outline affected the heaviness of the bottle and the ease of lifting. Red was heavier/more sluggish." (P10)

However, affective color associations alone do not fully explain the observed ordering. If perceived effort followed established color-emotion associations, we might have expected blue to yield lower effort ratings than red [27]. Instead, gray produced the lowest ratings. Participants' responses suggested that the visual salience of the outlines relative to the surrounding environment may have also contributed to this pattern. Although we purposely matched the luminance of all three outline colors to control brightness to reduce brightness-related differences, as brightness has been shown to modulate weight perception [41], participants tended to perceive the red outlines to be brighter, and gray outlines to not stand out as much:

"The red outline made me think the bottle was heavier and harder to lift. It was a much brighter color and I had to put more effort into lifting the bottles." (P3)

"Gray made it lighter since it didn't stand out as much. Red made me more cautious because of the usual beware energy that color has." (P19)

As shown in Figure 2, the perceived contrast between each outline and the surrounding scene may therefore have differed despite controlling their luminance. The main takeaway from this key finding is that red outlines increased perceived effort compared to gray outlines, but this effect may also be influenced by elements such as noticeability and contrast with the background, instead of only being a reflection of affective color associations.

Outline Thickness Did Not Reliably Alter Perceived Weight or Effort. Our hypothesis (H1) was that thicker hand outlines would increase the perceived weight of the bottles compared to thinner outlines. This hypothesis was not supported by the quantitative results. Although previous typography and packaging research indicates that bold, visually weighty forms can connote heaviness [23, 29, 32], this association did not produce a reliable group-level change in perceived physical weight or lifting effort.

The qualitative responses, however, revealed substantial variation between participants. Twelve participants (40.0%) explicitly associated thicker outlines with greater weight or effort:

“Generally, I associated thicker outlines with heavier bottles, while thinner outlines made objects feel lighter or easier to lift.” (P20)

Some participants located this effect in the appearance of their own body rather than solely in the bottle:

“The thickness made my hand feel heavier or lighter which made it easier or harder to lift.” (P19)

Conversely, 14 participants (46.7%) reported that thickness had no effect, frequently because they did not notice or differentiate the thickness levels during the interaction:

“I had no perception that the thickness changed, I didn’t even realize that could change.” (P29)

These responses indicate that the null group-level effect coexisted with a recognizable thick-heavy association for a substantial subset of participants. Rather than suggesting that outline thickness carried no heaviness-related meaning, our findings suggest that this association was heterogeneous and was not sufficiently salient or consistent across participants to reliably influence the quantitative ratings. Future work could apply thickness manipulations directly to the object to determine whether the association transfers more consistently when the visual cue and the judged physical property share the same perceptual target.

6.2 Applications to Various Domains

Our findings are applicable to design of embodied and tangible interaction in mixed and augmented reality. As lightweight head-worn displays move toward longer-term, pervasive use, they afford us the ability to couple physical objects with digital information [1, 30]. Correspondingly, users may interact with that information through embodied and tangible means (e.g., grasping, lifting, physical manipulation) [25]. Applying red outlines to hands may serve as a feedback cue to increase perceived effort while handling such objects to subtly discourage certain actions while they are being done, or to place emphasis or importance to them.

A particularly direct application is mixed and augmented reality tabletop gaming. Prior work has investigated both players’ responses to augmented tabletop games and systems that combine AR graphics with tangible or actuated tabletop objects [28, 31]. In this setting, a hand-anchored outline displayed through a head-worn device could provide player-specific feedback without physically modifying a shared game piece. For example, a red outline could appear when a player lifts a “cursed” artifact, moves an overencumbered character, pays a stamina cost, or attempts an action associated with resistance or danger. The same physical token could therefore acquire different effort-related meanings according to the player’s role, abilities, or current game state. The outline could also change from gray to red as fatigue, penalties, or difficulty accumulate, making these mechanics visible at the moment of manipulation.

Embodied education and simulation present another promising application area. Tangible AR systems have used manipulable tabletop objects to expose otherwise invisible forces, constraints, and collisions, including applications in physics education and mechanism exploration [28]. Together with prior research demonstrating that AR visual manipulations can influence perceived weight and

fatigue [5], our findings motivate the investigation of color-based effort cues in such experiences. In an educational simulation of friction or mechanical advantage, for example, the same constant-weight prop could represent different virtual loads, with a red hand outline identifying the state intended to require greater effort. Similar feedback might emphasize a mechanically inefficient grasp or a demanding step during procedural training. This cue could connect an abstract system state to the learner’s embodied action without requiring the physical load itself to change.

6.3 Limitations and Future Work

As stated in Section 6.1, our results may have also been influenced by the contrast between the table and the outline colors alongside heightened arousal. Although we controlled the luminance values of all three colors, perceived brightness is also dependent on contrast, which was not accounted for in our experimental design. For instance, if the table and surrounding environment were red, then we may not have observed the significant effect of line color on perceived effort. However, a separate user study would be required to confirm this.

Participants’ responses suggest that the physical feedback from lifting the real-world bottles may have outweighed the visual effects of the hand outlines. Since prior work has shown that hand outlines can influence physical sensations such as roughness, and our results suggest that outline color can affect perceived effort, future work could replicate this experiment in VR using fully virtual objects. Without physical weight cues, participants may rely more heavily on visual information, potentially allowing hand outlines to have a stronger effect on perceived weight.

Future work in modulating weight and effort perception while handling physical objects could involve other visual cues, such as particle effects. For instance, particles accelerating downwards, localized around the object being lifted, may induce a feeling of heaviness or downwards acceleration on the object itself. Similarly, particles that are warm-colored or cold-colored may induce temperature illusions. Manipulating latency of the AR hand avatars, and seeing if higher latency induces sluggish/heavy sensations may also be an interesting avenue to explore. Collecting physiological data as participants grip and lift objects, rather than subjective questionnaire responses, may also be an interesting approach for future work of this nature.

7 Conclusion

In this work, we investigated whether the color and thickness of AR hand outlines could influence the perceived weight of real-world objects and the effort required to lift them. Thirty participants lifted visually identical bottles of three different weights while viewing hand outlines that varied across three colors and three thicknesses. We found that outline color significantly influenced perceived effort, with red outlines producing greater effort ratings than gray outlines and blue outlines falling numerically between them. However, color did not significantly affect perceived weight, and outline thickness did not reliably alter either perceived weight or effort. Thus, our findings partially support the hypothesis that red hand outlines increase perceived effort, but do not support the hypothesis that visually heavier, thicker outlines increase perceived object weight.

These results demonstrate that a visual effect applied to the user's body can shape an embodied judgment while leaving both the appearance and physical properties of the manipulated object unchanged. At the same time, the separation between effort and weight suggests that hand-centered color cues may influence users' interpretation of their own exertion more readily than their perception of an object's physical mass. Red hand outlines may therefore provide a lightweight mechanism for communicating resistance, difficulty, urgency, or fatigue in tangible mixed reality applications such as tabletop games, education, and procedural training.

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